

The Bilateral and Population-Scale Governance Boundary

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2025

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Claim 6 of the third CKS paper projects that accumulated Full Aspect Integration (FAI) events across many coordinating Selves produce population-scale collective evolution. This raises a boundary question: at what point does a collection of bilateral FAI relationships cross into population-scale coordination? Does a threshold network size exist above which new governance mechanisms are required? This note resolves the boundary by establishing that no such threshold exists. Population-scale dynamics are an emergent property of accumulated Claims 1–5 governance events, not a separate governance mode that activates at scale. The governance architecture is identical at every scale — bilateral, small network, emerging network, and population scale — and Claim 6’s three sub-commitments are live as soon as a Self governs more than one FAI relationship. The prior-art significance is direct: there is no governance novelty at population scale that is not already present at bilateral scale. Any adversarial claim about population-scale AI coordination governance must address that the architecture covering it has been specified, at bilateral granularity, in the prior art. This note is written in the calibrated-humility register appropriate to Claim 6, reflecting that population-scale dynamics are projected consequences of the architecture rather than empirically verified observations.

1. The Boundary Question

Claim 6 of the third CKS paper articulates what the architecture’s five preceding claims compose into at population scope: accumulated FAI events across many coordinating Selves produce population-scale collective evolution, governed by joint authority, topology governance, and a convergence-diversity balance under human governance. This projection raises a definitional

boundary question that must be answered for the prior art to be precise.

Consider four network states:

Scenario (a) — One bilateral pair. Self A and Self B complete one FAI event. Claims 1–5 governance applies: shared substrate, conflict preservation, human-governed authority, evolution feed, configuration.

Scenario (b) — Small network. Self A has FAI relationships with Selves B, C, and D. Self B has relationships with A and C. Self C has relationships with A and B. Self D has a relationship only with A. Four Selves; seven bilateral relationships; no Self governs more than three FAI connections.

Scenario (c) — Emerging network. Twenty Selves in a shared domain each maintain three to five bilateral FAI relationships. No central coordinator; governance is distributed across bilateral pairs and small groups.

Scenario (d) — Population scale. Hundreds of Selves across many domains have extensive bilateral and small-group FAI relationships. The accumulated events are numerous; the network topology is complex.

The boundary question is: where does Claim 6’s population-scale scope begin? Is there a threshold N — some minimum number of Selves or FAI events — above which a different or additional governance architecture is required?

2. The Architecture’s Resolution: No Hard Threshold

The architecture resolves the boundary question cleanly: there is no hard threshold, and there is no governance architecture above the threshold that is distinct from the governance architecture below it.

Population-scale collective evolution is an **emergent property** of accumulated FAI events, not a separate governance mode. As more Selves engage in FAI relationships, and as those relationships accumulate over time, the aggregate of Claims 1–5 governance events produces network-level dynamics — convergence of shared substrate content across coordinating Selves, divergence and specialization where governance configurations differ, capability propagation as new Selves enter FAI relationships and inherit substrate content from established ones. These dynamics are what the architecture produces at scale. They are not produced by a new governance mechanism that activates when the network crosses a size threshold.

The governance architecture is **identical across all four scenarios**. In scenario (a), Self A governs one FAI relationship using Claims 1–5 governance. In scenario (d), each Self in the population governs its portfolio of FAI relationships using the same Claims 1–5 governance. No

new claim is required. No new architectural commitment is activated. The population-scale dynamics are the aggregate output of many bilateral and small-group Claims 1–5 events operating under the same design pattern.

This architectural continuity is the core prior-art claim. Any attempt to identify governance novelty at population scale must first address that the governance architecture across all four scenarios is Claims 1–5 applied to FAI events. There is no governance mechanism in scenario (d) that is not present, in its architectural essentials, in scenario (a).

3. Where Claim 6’s Sub-Commitments Apply

Claim 6 specifies three sub-commitments that formalize what Claims 1–5 governance composes into at the network position level. Identifying when each sub-commitment first applies is load-bearing for the prior-art scope.

Joint authority at population scope. Claim 6 projects that FAI relationships extend the bilateral joint-authority model of Claim 1 to whatever scope the network occupies. This sub-commitment is active at scenario (a): the moment Self A and Self B complete one FAI event under joint authority, the joint-authority architecture is in operation. Scenarios (b), (c), and (d) extend it across more relationships and more Selves; the architectural commitment does not change.

Topology governance. Claim 6 projects that a Self governing multiple FAI relationships must manage its network position as a portfolio — choosing which FAI relationships to initiate or maintain, what configurations to apply, how to prioritize evolution-feed across relationships. This sub-commitment first becomes live at scenario (b). Self A, with FAI relationships to B, C, and D, is already in the position of governing its topology: three active FAI connections, with distinct configurations possible for each. Topology governance is not a large-network concept; it is a multi-relationship concept, and it applies at scenario (b) with four Selves.

Convergence-diversity balance. Claim 6 projects that governance over accumulated FAI events involves a balance between convergence (shared substrate content propagating across coordinating Selves) and diversity (Selves preserving distinct governance configurations appropriate to their domains). This sub-commitment is conceptually present as soon as a Self has more than one FAI relationship — scenario (b) — because distinct relationships may feed different substrate content, and the governance configurations for each are independently specifiable. The balance becomes more observationally prominent at scenarios (c) and (d), where accumulated events are numerous enough that convergence and divergence patterns are visible across the network. But the architectural commitment is already in place.

The practical implication is that Claim 6 is not a “large-network” claim. It is the claim about governing any FAI network position — from the smallest multi-relationship scenario onward.

The population-scale framing in the source paper names what the architecture is projected to produce when many Selves operate under it, not the minimum size at which the architecture applies.

4. Population-Scale Dynamics as Emergent Consequence

The calibrated-humility register is appropriate here because the claim about population-scale dynamics is a projection, not an empirically verified observation. Population-scale deployments of the CKS-governed architecture do not yet exist. The claim is that if such deployments occur, the dynamics the architecture is projected to produce — convergence, divergence, specialization, capability propagation — follow from the architecture without requiring fresh theoretical apparatus at population scope.

This is the sense in which population-scale dynamics are emergent. The term is not used in the complex-adaptive-systems or irreducibility sense; no fresh emergence theory is required. The dynamics are emergent in the simpler sense that they are properties of the aggregate that are not visible at any single bilateral event but become visible when many such events accumulate. They follow from composition, not from new mechanism.

The architecture does not prescribe what the convergence-diversity balance looks like at population scope, nor does it prescribe what topology governance decisions any Self makes. These are deployment-level and governance-level decisions made by the humans exercising authority over their Selves' FAI configurations. What the architecture specifies is that those decisions are made using Claims 1–5 governance at every scale; the dynamics that result are the aggregate of those decisions.

5. Prior-Art Significance and Boundary Test

The prior-art significance of this boundary resolution is direct. The governance architecture covering population-scale coordination is Claims 1–5 applied to FAI events across many Selves. That architecture has been specified at bilateral granularity. Any adversarial claim about novel population-scale AI coordination governance must address that the complete architecture — shared substrate, conflict preservation, human-governed authority, evolution feed, configuration — is already present, and already prior art, at the bilateral level.

There is no governance novelty at population scale that is not present at bilateral scale. The architecture does not become more complex as the network grows; it becomes more numerous. Many instances of Claims 1–5 governance, operating across many bilateral and small-group FAI relationships, produce population-scale dynamics as a consequence. The consequence is prior art because the mechanism that produces it is prior art.

Boundary test. For any FAI coordination scenario, at any scale, the test is: does the coordination use Claims 1–5 governance? If yes, the scenario is within the prior-art scope at whatever scale it operates. There is no scale threshold above which a different governance architecture is required. Bilateral, small-network, emerging-network, and population-scale coordination scenarios all pass the same test, and all are within the prior-art scope.

This boundary test applies in the calibrated-humility register: it is a claim about what the architecture specifies and what the prior art covers, not a claim about empirically observed population-scale dynamics. The dynamics await observation. The architecture that governs them does not.

Derivation note #663 in the CKS defensive publication series. Phase D6, note D6.13.